Keating: A Metaharness for Agency-Preserving AI Instruction

Dio the Debugger

September 06, 2026

Revised architecture: evidence-gated teaching skill evolution

AI tutors can generate persuasive explanations without establishing that learners can reconstruct, retain, or transfer an idea. Keating addresses this measurement problem through a teaching metaharness: a control layer around live instruction, learner records, artifacts, and revisable teaching procedures. This revision replaces automatic promotion from policy-dependent proxy scores with fresh tutor executions, fixed rubric judgments, persistent hypotheses, and content-addressed skill revisions. An 18-case mathematics and programming suite separates training, validation, and a single-use holdout. A candidate must improve under both independent comparisons, preserve every case family's score, and pass all critical criteria before subsequent sessions may use it. Recorded quiz performance, feedback proxies, synthetic teaching behavior, and independent learner assessments remain distinct. Historical analysis explains the redesign: a frozen policy gains 3.982 points inside an algebraic score model, yet 30 standardized MAP-Elites reruns yield 11 improvements, four ties, and 15 regressions. The archived model-to-model traces also expose student-role contamination. These historical results are not evaluations of the new skill loop. The present contribution is an implemented method with reproducible integrity checks and explicit limits: no live-provider performance results for the new loop or human learning effects are reported. Judge calibration and randomized trials with delayed and transfer assessments remain necessary.

Introduction

One-to-one tutoring and intelligent tutoring systems provide important evidence that instructional support can improve learning (Bloom, 1984); (Ma et al., 2014). Recent work includes a randomized study of a structured generative-AI tutor (Kestin et al., 2025) and a deployed generative teaching assistant examined through use and learner perception (Thesen & Park, 2025). Such findings do not validate every system called an AI tutor. The instructional procedure, learner population, comparison condition, and outcome measure matter.

Keating's promise is to help learners reconstruct ideas and use them independently. Its teaching design emphasizes diagnosis, self-explanation, retrieval, and transfer, drawing on established research (Chi et al., 1989); (Rittle-Johnson & Loehr, 2017); (Roediger & Karpicke, 2006). Fluent explanations alone are an inadequate success measure: cognitive offloading can change what a learner does and subsequently remembers (Risko & Gilbert, 2016); (Burnett & Richmond, 2026). Conversely, requiring questions or retrieval in every situation can be unhelpful when a learner needs an explicit explanation. An agency-preserving tutor must respond to the learner's actual need, not maximize the visible frequency of a preferred teaching technique.

This paper describes Keating as a **teaching metaharness**: a controller around the teaching exchange that maintains artifacts, learner evidence, and rules for changing the tutor. Terminal and browser interfaces support live teaching alongside plans, maps, quizzes, decks, verification, and other activities. Their breadth makes the central evaluation question sharper: does a change improve what the tutor does, and does that improvement subsequently help a learner? Interface coverage, artifact generation, satisfaction, and delayed unaided performance answer different parts of that question.

An audit of the earlier improvement path identified three measurement problems. A deterministic simulator could reward changes to its own policy coordinates without executing changed teaching. Candidate-dependent weights made search scores difficult to compare. Historical learner feedback could describe an old interaction but could not supply the learner's counterfactual response to a new tutor. The historical results retained here illustrate these problems; they do not establish the efficacy of the replacement.

The revised system separates raw executions, accumulated hypotheses, and deployable skills. This organization is informed by WikiSkill's distinction between immutable experience, persistent knowledge, and procedural skills (Tang et al., 2026). Keating implements a smaller hypothesis ledger and one reflective proposal per invocation; it does not reproduce WikiSkill's full wiki-maintenance system or claim its reported task-performance gains. Related work on harness optimization provides the broader systems context (Lee et al., 2026); (Zhang et al., 2026).

The contributions are: (1) an implemented procedure for evaluating and activating teaching skill revisions against fixed behavior criteria; (2) explicit measurement boundaries between observed assessments, proxies, synthetic continuations, and human learning outcomes; and (3) a reproducible analysis connecting the earlier system's failure modes to the new design. Agency preservation is the instructional objective, not an empirically established property of this release.

Current System and Design Rationale

What changed

Keating now has three distinct evaluation entry points. `bench` summarizes recorded assessment and feedback evidence. `teaching-bench` executes new tutor continuations on training cases. `auto-improve` proposes a teaching skill and evaluates its exact revision on fresh paired comparisons before activation. The first operation cannot establish the effect of changing a tutor; the second can expose teaching behavior; the third makes a bounded decision about experimental use.

Legacy `evolve` still produces parameter proposals for inspection and research. It no longer installs an unvalidated policy through that command. Legacy `prompt-evolve` scores and snapshots remain diagnostics; the newest saved snapshot is not automatically the active teaching prompt. MAP-Elites and PROSPER-style selection remain available as research mechanisms, but they are not the new skill activation gate. A count of five feedback records is not a validation sample or evidence of improvement.

Runtime, evidence, and revision state

The live runtime uses Pi in the terminal and a Pi Agent in the browser. It combines a base teaching prompt or selected persona with teaching tools and learner context. Local deterministic functions continue to produce inspectable pedagogical artifacts. The new shared experiment engine in `shared/evolution/` controls the evaluator, corpus, proposal, and decision independently of those live interfaces.

The experiment state has three complementary parts. **Raw evidence** stores the fixed case manifest, actual messages and tool calls, criterion judgments, errors, model/runtime attribution, and unique execution identifiers. **Hypotheses** record a proposed explanation or teaching adjustment, its evidence references, and whether it remains proposed, was supported offline, or was rejected. **Revisions** bind the base prompt and complete skill contents to a SHA-256 identifier, a parent revision, and an experiment record. Rejecting a candidate preserves its evidence and hypothesis; it leaves the incumbent active.

Earlier failure mode	Implemented response	Remaining evidence gap
A policy change alters a retrospective proxy score	A fixed recorded corpus has policy- and weight-independent scores	Observed performance is not a causal treatment effect
A prompt diagnostic or synthetic policy score authorizes use	Only a validated skill revision can enter the new activation path	Rubric judgments still need human calibration
Evidence used for search appears to validate the result	Training informs proposals; validation and a consumed holdout gate activation	Public cases and semantic overlap can contaminate evaluation
Discarding an edit also discards its rationale	Evidence-linked hypotheses persist after rejection	Long-run benefits of accumulated hypotheses remain unmeasured
A changed prompt inherits an old score or changes a live session	Content-addressed revisions and session pinning bind evaluation to instructions	The bounded evaluator does not cover every live tool or interface

Table 1: The redesign addresses specific evaluation defects without converting implementation safeguards into efficacy claims.

The deployed teaching procedure receives the base prompt and active skill instructions. It does not receive the hypothesis ledger, rubric, or held-out evaluation results through the evaluator interface. The proposer receives training evidence and retained hypotheses. A separate judging call receives the case rubric and actual execution, without the candidate's skill instructions. These are role and information boundaries; the default adapters can use the same underlying provider and model for all roles.

Before activation, the store saves the complete decision record and checks the active pointer under an exclusive lock. Later activation and resumed-session loading revalidate the revision and its saved evaluation evidence. Existing sessions retain their prompt revision; new sessions can use an accepted revision with the matching base persona. Changing the base prompt starts from that base's own empty skill revision instead of inheriting an unrelated evaluation result.

CLI/Pi records live under the current project's `.keating/state/teaching-evolution/`. Browser records use a separate IndexedDB database and Web Locks. The browser evaluator uses disposable storage; its evaluation and activation modules are excluded from the mutable NodePod boot bundle. This is a local trust boundary, not protection from a malicious owner rewriting the application or its database. Durable state is currently project-local or browser-origin-local, with no account-wide synchronization and no browser account namespace.

Bounded teaching execution

Evaluation invokes the actual pedagogical runtime with a restricted tool set. A CLI episode starts a fresh Pi child process and temporary workspace, with in-memory sessions and no ambient project context or user extensions. Allowed tools are `plan`, `map`, `verify`, `quiz`, `grade_quiz`, and workspace-confined `read`. The browser adapter creates a fresh Pi Agent and disposable IndexedDB database with `deck`, `quiz`, `grade_quiz`, and `grade_question_checks`. It uses the selected model and thinking level. Shell execution, source edits, recursive evolution, and animation's nested inference are excluded from these evaluators.

The default episode limits are 90 seconds, six provider calls, eight tool calls, and 2,048 output tokens per provider call. CLI subprocess output is limited to 1 MiB; browser output is limited to 128,000 characters. Timeout and cancellation reach the running actor, and disposable workspaces are cleaned up. The judging and proposing adapters have no tools. These controls make experiments bounded and inspectable, but do not reproduce the full live environment: speech, arbitrary sandbox tasks, courses, collaboration, deployed authentication, and UI usability require separate evaluation.

Methods

Evidence types and the estimand

The revised method asks whether a specified change improves **tutor behavior on fixed cases** before permitting experimental teaching use. It does not estimate learning gain from the tutor's prose. Let R denote a revision, x_i a learner conversation prefix, and $\tau_{i(R)}$ a freshly executed tutor continuation, including any permitted tool use. A fixed rubric J_i assigns binary criterion judgments to that execution. The per-case score is

$$q_{i(R)} = \left(\frac{1}{m_i}\right) \sum_{j=1}^{m_i} J_{ij}\left(\tau_{i(R)}\right),$$

where m_i is the number of criteria for case i. A favorable aggregate cannot compensate for a failed critical criterion. Each criterion includes a rationale in the saved evidence.

The cases stop at a learner turn requiring an instructional decision. The actor generates the subsequent tutor continuation; the benchmark does not append a historical learner answer as if it were a response to the new teaching. It also does not generate an adaptive simulated learner for subsequent turns. The resulting score measures the next teaching decision under a synthetic prefix, not learner uptake, a complete instructional session, or delayed learning.

Retrospective `bench` data have a different meaning. Recorded quiz scores are observed assessment performance. Explicit sentiment and inferred learner-turn signals are labeled proxies. Changing a candidate policy or objective weights cannot change the score assigned to the same recorded corpus. Without independent measurements, mastery gain, retention, and transfer remain unavailable. Legacy numeric compatibility fields may contain zero, but evidence metadata and reports distinguish missingness from a measured incorrect answer. Neither sentiment nor an algebraic policy bonus is converted into retention.

Case construction and information separation

The bundled `teaching-behavior-v1` suite contains 18 curated cases across mathematics and programming. Each split has six distinct family labels, with three cases from each domain. A family cannot appear in multiple splits. Cases include concrete misconceptions and explicit learner requests, with criteria for subject correctness and the requested instructional act. For example, a learner may need one diagnostic question, a hint without the final answer, or a direct explanation. This prevents a single visible behavior such as asking more questions from defining success everywhere.

Training cases supply examples for reflection. Validation compares incumbent and candidate after the proposal is fixed. A release holdout is consulted only after validation passes. The actor does not receive rubric contents; the proposer does not receive validation or holdout conversations or judgments. The fixed judge sees the actual continuation and its rubric, not the revision's instructions. Separate calls reduce direct leakage without making the judge statistically independent of the actor or immune to persuasion in a response.

The suite and held-out content are public repository material. "Sealed" therefore means withheld from the automated proposal path and tracked for consumption; it does not mean private from developers or absent from model pretraining. Distinct family names are an enforceable schema condition, not proof of semantic or statistical independence. Independent authors must supply genuinely new tasks when renewing the release evidence.

Reflective proposal and paired gate

Each invocation executes the incumbent on training cases, then requests one skill creation or replacement with an explicit hypothesis and references to training evidence. It preserves earlier hypotheses, including rejected ones, so subsequent proposals can use accumulated observations. The current proposer is a bounded reflective call. It does not train model weights, implement a full GEPA optimizer, or revise the evaluator, scoring thresholds, permissions, or source code.

For each evaluation split, the incumbent and candidate execute on the same fixed cases and repeat count. All case/repeat pairs must be present. Their model and runtime identities must match. Each run has a fresh identifier, and saved aggregate scores are checked against the underlying judgments before they can authorize activation. Failed actor or judge calls produce explicit errors and a null aggregate; the gate never averages only the successful survivors or substitutes a deterministic score.

Let d_f be the mean candidate-minus-incumbent score over all cases and repeats in family f. Families receive equal weight in the comparison, so additional variants or repeats do not count as additional independent families. With F families, the reported effect is $\overline{d} = \left(\frac{1}{F}\right) \sum_{f=1}^{F} d_f$. Let w and l count positive and negative family differences; ties within a numerical tolerance of 10^{-9} are excluded from the sign test. The one-sided tail probability is

$$p = \sum_{k=w}^{w+l} \binom{w+l}{k} 2^{-(w+l)}.$$

When there are no non-tied families, the implementation assigns $p = 1$. Acceptance requires all of the following on **both** validation and holdout:

- at least six distinct paired families;
- a mean improvement of at least 0.05 on the 0-1 rubric scale;
- no family regression beyond the numerical tolerance;
- no failed critical criterion in any candidate execution;
- $p \leq 0.05$, complete evidence, and compatible execution provenance.

These thresholds are fixed engineering eligibility rules, not an empirically calibrated guarantee of better teaching. With six families and no losses, five wins and one tie give $p = 0.03125$; four wins and two ties give $p = 0.0625$ and fail. The statistical interpretation depends on independent, exchangeable family differences under the null. Authored case labels, shared subject matter, correlated model errors, and repeated proposal selection can violate that assumption. The two gates do not create a human-outcome confidence interval or establish a global false-promotion rate across a lifetime of experiments.

Experiment lifecycle and holdout accounting

```
lock project or browser evolution state
verify incumbent revision and previous activation evidence
check budget, cooldown, and unused holdout content/families
run fresh incumbent training continuations
save one evidence-linked hypothesis and candidate revision
run paired incumbent/candidate validation
if validation passes:
    persist holdout consumption before any holdout execution
    run paired incumbent/candidate holdout
save raw evidence and accepted/rejected/failed experiment
if both gates pass:
    atomically activate the evaluated candidate revision
retain hypotheses and preserve existing session revisions
```

The bundled suite requires 30 tutor executions for a complete run at one repeat: six training executions, 12 validation executions, and 12 holdout executions. Each completed tutor execution also requires a judging call, and the run has one proposing call. This is a planned upper path for the default suite, not a measured runtime or monetary cost. The loop allows at most two repeats and 60 tutor executions, including both revisions. Earlier failure or validation rejection stops the dependent stages.

A 30-minute cooldown limits repeated invocations. `--force` bypasses only that cooldown. Once the holdout is accessed, its fingerprint and family labels remain consumed after acceptance, rejection, or a crash. Reordering cases or changing record identifiers cannot reopen it. A later eligible invocation needs a genuinely new holdout pack; the system does not automatically manufacture independent evidence. CLI users can supply a bounded JSON case pack. Browser experiments currently use the bundled pack and need an application update to renew it. This is controlled, repeated skill evolution with an external evidence-supply dependency, not unlimited autonomous improvement.

Independent learner assessments

The separate learner-check bank currently covers fractions and loop bounds. Each topic has three distinct items at each of four stages: a precheck, immediate postcheck, delayed recall, and transfer. Numerical answers are graded by fixed code rather than an LLM. A delayed stage opens one day after the immediate submission and the transfer stage opens seven days after it; actual timestamps preserve late participation. The transfer stage does not require completing delayed recall.

Each record identifies the tutor revision selected at start. Assistance is recorded as `none`, `assisted`, or `unknown`. Only stages marked `none` contribute to unaided summaries. If both precheck and immediate scores are eligible, their difference is reported as descriptive immediate gain; eligible delayed and transfer scores are reported separately. Missing or ineligible stages remain null. Exact repeated submissions are idempotent, conflicting resubmissions fail, and persisted grades are checked by replay through the independent grader. Learner-facing prompts and records omit answer keys.

The bank is fixed and small. Its forms are not psychometrically equated, revision exposure is operator-recorded, and assistance is self-reported. The pre/post difference therefore cannot isolate instruction from item difficulty, testing effects, selection, or outside help. These CLI records provide a basis for future trials; they do not currently drive automatic skill promotion or establish a causal effect.

Historical archive and simulator protocol

For continuity, we retain the earlier diagnostic analyses. The archive contains 22 model-generated teaching traces. Selecting the latest timestamp for each topic and learner-model pair retains 16 sessions across four topics and four learner models, matching `test/final_dataset.json`. One record uses a 0-10 encoding rather than 0-1 and is divided by ten before aggregation. The mean of the archived mastery, engagement, and clarity labels is a descriptive composite. Scorer identity and inter-rater reliability are unavailable. Reported 95% bootstrap intervals use 5,000 resamples with deterministic seeds; they describe variation in this archive and do not repair its selection or labeling limitations.

The legacy simulator maps nine policy controls and eight sampled learner attributes to fit and overload terms, then to algebraic mastery, retention, engagement, transfer, and confusion values. It executes no revised tutor response. Its coefficients are available in `shared/pedagogy/benchmark-real.ts`; the Node module under `src/core/` exposes that implementation. In abstract form, its score is $S = \sigma_M M + \sigma_R R + \sigma_E E + \sigma_T T - \sigma_C C$. The outcome names designate simulated quantities, not measured human variables. Inspectability makes the mechanism reproducible while also making its preferred policy coordinates available to the optimizer.

The frozen `me-candidate-33` policy originated in Keating 3.3.0 and remains at `docs/study/evaluated-policy.json`. We reevaluate it and `DEFAULT_POLICY` with fixed default weights over seeds 1-200, 14 topics, and three pseudo-learners per topic, with provider judging disabled. One-at-a-time coordinate swaps use the same seeds. Thirty derivative-focused MAP-Elites runs use 24 candidates each and distinct temporary grids. Candidate search co-evolves weights and uses candidate-specific seeds; reported deltas reevaluate each selected policy against default on the same run seed with fixed weights. These analyses diagnose the legacy score model and search procedure; they are not runs of the new teaching-skill loop.

Reproducibility and implementation checks

The analysis script records the executed package version separately from the historical policy's origin version. It hashes a manifest of the relevant current source, package/lock files, base teaching prompt, suite, archive, and frozen policy. Package version alone does not identify the revised working tree. Generated JSON and Markdown include the case inventory, all historical seed comparisons, ablations, and isolated search reruns. They explicitly record that new-loop live-provider and human-learning results were not collected for this paper.

Local tests exercise fixed-corpus score invariance, evidence completeness, critical failures, revision binding, holdout reuse, concurrent activation, session pinning, and assessment timing and missingness. Runtime fixtures invoke actual Pi and browser Agent tool paths in disposable environments with controlled local responses. Passing those checks establishes behavior under the tested fixtures. It does not measure external model quality, judge agreement, teaching latency in production, or human learning effects.

Results

Current implementation inventory and evidence status

The current suite contains 18 cases with 18 distinct family labels. These are authored test inputs, not completed model runs or enrolled learners. The generated inventory validates their schema and separation; it does not estimate tutor performance.

Split	Cases	Math / code	Criteria	Critical criteria
Training	6	3 / 3	24	8
Validation	6	3 / 3	24	6
Holdout	6	3 / 3	24	7

Table 2: Current teaching-behavior suite, generated from the implemented cases. Each split has six distinct families; family labels alone do not prove independence.

The accompanying implementation checks cover malformed and missing judgments, paired omissions, modified cached scores, critical failures, stale revision references, holdout reuse, interrupted runs, concurrent activation, and delayed assessment rules. Controlled local runtime fixtures exercise real Pi and browser Agent tool execution. No live-provider score, acceptance rate, cost comparison, judge agreement, or human-learning gain is available for the new loop in this revision. Its reported status is an implemented and locally checked method awaiting those evaluations.

Historical evidence retained for diagnosis

The quantitative performance results below use two historical evidence layers, separate from the current inventory and implementation checks.

1. **Archival model-trace evaluation.** We analyze model-generated teaching traces stored under `test/traces/`. These are not human-participant sessions.
2. **Legacy deterministic benchmark.** We compare policies with the synthetic fallback implemented in `shared/pedagogy/benchmark-real.ts` and orchestrated by `src/core/benchmark.ts`.

The layers answer different questions. The archival layer asks which concrete successes and failures appear in recorded model-to-model teaching sessions. The synthetic layer asks how policies and the current search procedure behave inside an explicit score model. Neither layer estimates a causal effect on human learning.

The repository contains 22 raw trace files, including repeated runs for some topic x learner pairs. We imposed a deterministic curation rule: retain the latest trace by timestamp for each pair. This yielded 16 retained sessions, matching the versioned snapshot `test/final_dataset.json`. The timestamp rule fixes the set by temporal provenance rather than selecting whichever repeated run looked best. One retained derivative trace for `Qwen-2.5-1.5B` contained `mastery=8`, `engagement=7`, and `clarity=8` while the rest of the archive used a 0-1 scale; we normalized that record to 0.8, 0.7, and 0.8 and recorded the correction in the generated bundle.

Component	Value	Interpretation
Raw archived traces	22	All preserved model-generated teaching transcripts before curation
Retained topic x learner pairs	16	Archival evaluation set used in this paper
Excluded duplicate earlier runs	6	Older runs for the same topic x learner pair
Score corrections	1 record	Single 10x encoding error normalized before aggregation
Synthetic topics	14	Internal benchmark tasks implemented in code
Synthetic learners per topic	3	Deterministic pseudo-learners sampled per topic and seed
Frozen policy origin	Keating 3.3.0	Historical policy, reevaluated by the source manifest supplied with this revision

Table 3: Evidence layers, implementation snapshot, and curation rules.

Archival performance is heterogeneous across topics and learners

After normalization, the mean archival overall score, defined as the unweighted mean of mastery, engagement, and clarity, was 0.61 with a 95% bootstrap interval of 0.515-0.705. Performance varied sharply by topic. **Special Relativity** was strongest at 0.75 (0.596-0.883), followed by **Derivative** at 0.654 (0.454-0.767), **Social Contract Theory** at 0.613 (0.500-0.763), and **Stoicism** at 0.425 (0.283-0.558).

This pattern is substantively useful but not causal. The physics and calculus traces are structurally friendly to prediction, worked examples, and misconception repair, whereas the Stoicism traces demand introspective application. In this small archive, high instructional clarity does not reliably translate into demonstrated learner uptake on that introspective task.

Topic	n	Overall	Mastery	Interpretation
Special Relativity	4	0.750 (0.596-0.883)	0.695 (0.570-0.815)	Highest archived labels in this small set
Derivative	4	0.654 (0.454-0.767)	0.600 (0.263-0.788)	Variable model-role uptake in calculus traces
Social Contract Theory	4	0.613 (0.500-0.763)	0.537 (0.462-0.650)	Mixed archived engagement and mastery labels
Stoicism	4	0.425 (0.283-0.558)	0.287 (0.175-0.463)	Explanation often exceeds demonstrated learner uptake

Table 4: Archival model-trace evaluation by topic. Values are means with 95% bootstrap intervals.

Learner-model heterogeneity was also substantial. `Qwen-2.5-1.5B` scored highest overall at 0.779 (0.675-0.867), whereas `Llama-3.2-1B` scored lowest at 0.458 (0.367-0.550). We do not interpret these as broad claims about model families. The set is too small and the roles too artificial. The narrower result is that the teaching protocol was not uniformly robust across the four simulated learner profiles in the archive.

Student-role contamination is a central failure mode

The traces reveal a failure mode for agency-preserving instruction: student-role contamination. In some sessions, student turns begin to speak like a teacher or assistant rather than reconstructing the concept as a learner. Using a versioned heuristic over student turns, 5 of the 16 curated sessions showed at least one contamination marker.

These sessions performed worse descriptively. Sessions without contamination had mean mastery 0.575 and mean overall score 0.642, compared with 0.430 and 0.540 for contaminated sessions. The sample is small and the heuristic is not a validated classifier. Nonetheless, it demonstrates the kind of failure a metaharness can make visible: fluent text can coexist with weak evidence that the learner owns the idea.

The frozen policy is robust inside the legacy score model

The evaluated policy vector, `me-candidate-33`, is frozen in `docs/study/evaluated-policy.json`; the analysis does not read mutable `.keating` state. We compared that historical vector with the repository's default policy under the same default objective weights. Regeneration from this revision's source manifest reproduces the earlier archive and simulator results. This is a robustness comparison inside the legacy harness, not a held-out generalization result: the frozen candidate originated in full-suite MAP-Elites work.

Across 200 seeds, the frozen policy improved the default by 3.982 points on average (2.5th-97.5th percentiles: 3.039-4.985), winning on 200/200 aggregate suite comparisons. Its mean score was 59.039, compared with 55.057 for the default. Mean per-topic gains ranged from 3.142 points on **Legal Precedent** to 4.315 on **Cognitive Bias**; each topic improved on 196-200 of the 200 seeds.

The optimizer stability result is substantially more qualified. Thirty independent derivative-focused MAP-Elites runs used 24 candidates each and a fresh grid per run. Search candidates co-evolved policy coordinates and objective weights and were sampled on candidate-specific seeds, so their raw search scores are not directly comparable with the baseline. We therefore reevaluated each PROSPER-selected policy and the default policy on the same run seed with `DEFAULT_WEIGHTS`. Under that standardized comparison, 11 selections improved, four selected the unchanged default, and 15 regressed. The mean delta was -0.014, the median -0.164, and the observed range -8.805 to $+7.142$. This protocol finds diverse candidates, but it does not show reliable scalar improvement under a fixed objective.

Synthetic analysis	Result	n	Interpretation
Frozen policy vs. default	+3.982 points (3.039-4.985)	200 seeds	Aggregate suite delta is positive on every sampled seed
Per-topic robustness	+3.142 to +4.315 mean	14 topics	Each topic is positive on 196-200 seeds
Independent derivative MAP-Elites reruns	11 wins, 4 ties, 15 regressions; mean -0.014	30 runs	Standardized reevaluation shows no reliable scalar gain

Table 5: Reproduced legacy synthetic results using the policy frozen from Keating 3.3.0. These are not teaching-skill experiment results.

Ablations show what the legacy score model rewards

The frozen policy differs from the default in all nine controls. One-at-a-time swaps show that the largest gains come from lowering challenge rate (+2.227 points), maximizing retrieval practice (+1.820), and increasing interdisciplinary bias (+1.182). Increasing diagram bias to the frozen policy's maximum produces the largest negative isolated effect (-1.583), while reflection bias contributes −0.376 in isolation.

These are properties of the score model and the tested policy coordinates, not estimates of pedagogical effect. The result suggests that the legacy synthetic surface is especially sensitive to overload control and retrieval. It also shows why joint search can be hard to interpret: a parameter that hurts in isolation may interact with other coordinates or objective weights inside a selected policy. An optimizer that can see this surface may exploit those sensitivities without producing better human teaching.

Parameter swapped into default	Mean synthetic delta	Interpretation
challengeRate	+2.227	Lower challenge reduces modeled overload most strongly
retrievalPractice	+1.820	The harness rewards enforced recall
interdisciplinaryBias	+1.182	The simulator rewards this coordinate
analogyDensity	+0.657	Lower analogy density is weakly positive with a wide empirical range
exerciseCount	+0.343	Two rather than three exercises helps modestly
socraticRatio	−0.118	This coordinate is nearly neutral in isolation
formalism	−0.189	Additional formalism is slightly negative in isolation
reflectionBias	−0.376	Reflection emphasis is not rewarded in isolation
diagramBias	−1.583	Maximum visual emphasis is actively penalized in isolation

Table 6: One-at-a-time ablations diagnose the synthetic objective rather than human learning.

Discussion

How well does Keating fulfill its promise?

The strongest present claim is architectural: Keating implements a broad teaching environment and an auditable procedure for changing some of its teaching instructions. Its tools can organize instruction, elicit work, grade specified answers, and retain evidence across sessions. The new loop also makes a proposed revision execute before it can be accepted. These are concrete capabilities. They support evaluating Keating as a serious teaching system, but do not support a numerical rating of educational effectiveness from repository inspection alone.

The evidence remains weakest where the product promise is most consequential: whether learners gain durable, transferable understanding while relying less on the tutor. The archive contains model-role transcripts, and the legacy simulator directly maps favored policy controls to simulated outcomes. Neither supplies an estimate for that promise. A useful assessment therefore has three parts: implementation and traceability are demonstrable; teaching behavior is now testable under a fixed protocol but lacks new live-provider results here; human learning efficacy remains unmeasured. Combining these into one readiness score would conceal the missing evidence.

The old optimizer results sharpen that assessment. The frozen policy wins all 200 aggregate comparisons inside the algebraic surface, while standardized reevaluation of 30 later selections yields 15 regressions and a mean delta near zero. Those findings are compatible: a favorable point on a known score surface does not validate the selection procedure or its educational interpretation. The redesign removes co-evolved scoring weights and retrospective policy bonuses from the activation decision. It then tests a changed procedure through actual tutor execution rather than asking the same fixed observations to justify a counterfactual improvement.

What self-evolution means in this implementation

Keating's current unit of adaptation is a bounded teaching skill. The system retains raw observations and evidence-linked hypotheses, proposes one change, tests it, and advances an active revision only when both gates pass. Failed candidates can still contribute to later reflection. This makes the accumulated history useful independently of whether the latest edit succeeds.

This design does not yet establish that the history causes better future proposals. A hypothesis ledger can accumulate mistaken explanations, repeated strategies, or domain-specific rules that do not generalize. Its relationship to raw evidence makes those errors inspectable, not impossible. The ledger is also smaller in scope than a maintained knowledge base with contradiction resolution, retrieval, and consolidation. No autonomous code rewrite, model-weight training, cross-account learning, or indefinite holdout recycling is part of the new activation loop.

The key practical constraint is independent evidence supply. A consumed release pack must be replaced by new tasks, including after a rejected candidate. That slows iteration but prevents repeated consultation of the same release evidence through the ordinary API. An unattended service would need a controlled task-authoring process, a separate evaluator owner, explicit resource budgets, and policies for drift and evidence renewal. These are future operational components, not features established by the present implementation.

A stronger benchmarking program

The next behavior study should freeze cases and judging prompts before search, calibrate judgments against blinded human ratings, and evaluate the same underlying tutor model under four conditions: base instructions, a fixed expert-authored skill, proposals based only on the latest traces, and proposals with the accumulated hypothesis ledger. Equalize permitted tools and budgets. Use several independently authored packs and record all attempted proposals, errors, critical failures, family-level differences, latency, token usage, and cost. This would test whether evolution improves behavior and whether retained hypotheses add value beyond another attempt at prompting.

Adversarial cases should challenge the intended mechanism directly: generic question spam, inappropriate refusal to explain, copying answers instead of eliciting work, needless tool calls, false claims of verification, and instructions in a tutor output that try to influence its judge. The existing cases already vary hint and explanation requests, while the local tests exercise missing and altered evidence. A dedicated adversarial behavior study has not been completed. Matching actor and judge models, repeating evaluation in a fixed order, and reusing authoring conventions can all produce correlated errors even with separate splits.

A human study should follow with randomized assignment to a frozen Keating revision and a strong tutor baseline using comparable model access and instructional time. Prespecify the primary outcome, analyze all assigned participants with disclosed missingness, and separate assisted practice from unaided testing. Include baseline knowledge, immediate performance, delayed recall, transfer to new problems, and an explanation that requires learners to reconstruct the concept without the tutor. The new assessment records help implement that workflow, but their tiny fixed banks require expansion, equivalent-form work, and independent grading of richer responses before they can support such a study. A priori sample-size planning must use a stated meaningful effect and variance assumptions; the 18 synthetic cases are not that calculation.

The educational objective should remain independent reconstruction and useful transfer, with correctness and learner requests as constraints. Engagement, conversational fluency, session length, and the frequency of retrieval prompts can diagnose experience, but should not become interchangeable substitutes for learning. Spaced review remains motivated by distributed-practice research (Cepeda et al., 2006); Keating's scheduling heuristic is not calibrated or evaluated by this paper.

Limitations

No new efficacy evidence. No human participants or live-provider performance results from the new skill loop are reported. The 16 archived model-role sessions and algebraic simulations describe the earlier evaluation regime. Local test success establishes specified software behavior, not better teaching or deployment reliability.

Limited behavioral coverage. The starter suite has 18 cases in two domains and evaluates the tutor's next continuation after a fixed learner prefix. It cannot observe a learner's reaction to changed teaching or test a full sequence of diagnosis, instruction, repair, and return visits. The CLI and browser allow different pedagogical tool subsets. Neither evaluator reproduces every model setting, tool, context source, voice mode, or UI state available to a live learner. Revision pinning controls instructional text; it does not freeze an external model's implementation or all surrounding product behavior.

Judge validity and provenance. Rubric judgments can favor style, verbosity, or familiar instructional language. The same model may act and judge in separate calls, and a transcript can contain attempts to influence evaluation. The adapter asks judges to treat transcripts as untrusted evidence, but that instruction is not a proof against manipulation. Saved records identify tutor model/runtime and criterion rationales; they do not yet provide a complete independently versioned judge configuration, calibration dataset, and inter-rater agreement study. Blinded human calibration remains necessary.

Small and public holdouts. Distinct family labels do not guarantee independent errors. The release cases are public source, not private evaluation material. Consumption records block ordinary reuse but cannot detect every semantic paraphrase or prevent a privileged developer from rewriting the store. The sign-test rule assumes conditions that the small authored suite has not empirically established; it is an eligibility check, not a lifetime error-rate guarantee. New holdout packs require independent authorship, and the browser has no operator-facing pack-renewal workflow yet.

Incomplete longitudinal measurement. Learner checks cover two topics with three items per stage. Parallel forms are not validated or equated, assistance is self-reported, and revision exposure is recorded by the operator. Missingness is explicit but may be systematic. No mechanism currently randomizes learners, controls outside help, or attributes a pre/post difference causally to a revision. Learner outcomes do not yet enter automatic promotion decisions.

Bounded and local adaptation. Only skill instructions evolve through the new gate. The hypothesis ledger has no demonstrated long-horizon consolidation benefit. CLI and browser state are separate; browser state has no account namespace or cross-device synchronization. A changed base persona needs its own evaluation. These constraints limit the scope of claims about a system that continually learns from all use.

Historical evidence quality. The archive is small, selected by a reproducible timestamp rule rather than a representative sampling design, and lacks scorer provenance. Its contamination detector is heuristic. The simulator's outcome names denote engineering equations, and its optimizer can exploit those equations. Keeping those analyses reproducible does not make them valid measurements of mastery, retention, or agency.

Code and Reproducibility Availability

Keating is available under the Mozilla Public License 2.0 at github.com/Diogenesoftoronto/keating. This September 6, 2026 revision describes the implementation in release 3.13.0 and its generated source manifest. Use the matching release tag and manifest hashes to identify the evaluated source; older releases do not contain the new implementation. The frozen policy's historical origin remains version 3.3.0.

The manuscript entry point is `docs/study.typ`. Current experiment contracts, cases, fixed scoring, hypotheses, skill revisions, and assessment logic are in `shared/evolution/`; runtime adapters and durable stores live under `src/core/`, `src/runtime/`, and `web/src/keating/`. `docs/teaching-evolution.md` documents commands, budgets, case-pack renewal, activation, and local storage scope. The analysis script and generated artifacts contain the exact source manifest and suite digest; manuscript inventory values are read from that generated JSON.

Run `devenv tasks run keating:study-analysis` to regenerate `docs/generated/study-analysis.json` and its Markdown companion. Run `devenv tasks run keating:paper` to regenerate the analysis and compile `web/public/keating-metaharness.pdf`. `keating:paper-check` compiles to a temporary PDF without replacing the web asset. These paper commands disable optional provider judging and perform no live-model experiment. Compare generated payloads without `generatedAt` for a deterministic rerun; compare their source manifests before attributing changed results to a method.

The root and browser test tasks are `keating:test` and `keating:test-web`. Focused fixtures include observed-benchmark, teaching-cases, teaching-evolution, teaching-prompt, teaching-episode-runner, and learning-check tests. The browser additionally checks disposable stores and persona/revision loading. These tests can run with local fixtures independently of provider access.

Data Availability

The 22 raw model-generated traces remain under `test/traces/`, the 16-record curation snapshot is `test/final_dataset.json`, and the historical policy is frozen in `docs/study/evaluated-policy.json`. Generated analysis records the curation manifest, score correction, archive summaries, 200-seed policy comparisons, ablations, and 30 isolated legacy optimizer reruns. It also records the current 18-case suite inventory, source hashes, and the absence of new-loop live-provider and human-learning results in this paper.

No private learner records or account credentials are inputs to the reproducible analysis. Future actual experiment traces and learner checks are stored separately in the project or browser, and require their own consent, data-management, and reporting decisions before research use.

Bibliography

Bloom, B. S. (1984). The 2 Sigma Problem: The Search for Methods of Group Instruction as Effective as One-to-One Tutoring. *Educational Researcher*, *13*(6), 4–16. https://doi.org/10.3102/0013189X013006004

Burnett, L. K., & Richmond, L. L. (2026). Meta-analytic Investigations of the Effect of Cognitive Offloading on Memory-based Task Performance and Interindividual Variability. *Memory & Cognition*, *54*(1), 144–168. https://doi.org/10.3758/s13421-025-01743-8

Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). Distributed Practice in Verbal Recall Tasks: A Review and Quantitative Synthesis. *Psychological Bulletin*, *132*(3), 354–380. https://doi.org/10.1037/0033-2909.132.3.354

Chi, M. T. H., Bassok, M., Lewis, M. W., Reimann, P., & Glaser, R. (1989). Self-Explanations: How Students Study and Use Examples in Learning to Solve Problems. *Cognitive Science*, *13*(2), 145–182. https://doi.org/10.1016/0364-0213(89)90002-5

Kestin, G., Miller, K., Klales, A., Milbourne, T., & Ponti, G. (2025). AI Tutoring Outperforms In-Class Active Learning: an RCT Introducing a Novel Research-Based Design in an Authentic Educational Setting. *Scientific Reports*, *15*, 17458. https://doi.org/10.1038/s41598-025-97652-6

Lee, Y., Nair, R., Zhang, Q., Lee, K., Khattab, O., & Finn, C. (2026). Meta-Harness: End-to-End Optimization of Model Harnesses. *Arxiv Preprint Arxiv:2603.28052*. https://doi.org/10.48550/arXiv.2603.28052

Ma, W., Adesope, O. O., Nesbit, J. C., & Liu, Q. (2014). Intelligent Tutoring Systems and Learning Outcomes: A Meta-Analysis. *Journal of Educational Psychology*, *106*(4), 901–918. https://doi.org/10.1037/a0037123

Risko, E. F., & Gilbert, S. J. (2016). Cognitive Offloading. *Trends in Cognitive Sciences*, *20*(9), 676–688. https://doi.org/10.1016/j.tics.2016.07.002

Rittle-Johnson, B., & Loehr, A. M. (2017). Eliciting Explanations: Constraints on When Self-Explanation Aids Learning. *Psychonomic Bulletin & Review*, *24*(5), 1501–1510. https://doi.org/10.3758/s13423-016-1079-5

Roediger, H. L., & Karpicke, J. D. (2006). Test-Enhanced Learning: Taking Memory Tests Improves Long-Term Retention. *Psychological Science*, *17*(3), 249–255. https://doi.org/10.1111/j.1467-9280.2006.01693.x

Tang, L., Rashtchian, C., Ferng, C.-S., Tomkins, A., Juan, D.-C., & Vu, T. (2026). WikiSkill: Compiling Agent Experience into Persistent Knowledge for Skill Evolution. *Arxiv Preprint Arxiv:2608.27454*. https://doi.org/10.48550/arXiv.2608.27454

Thesen, T., & Park, S. H. (2025). A Generative AI Teaching Assistant for Personalized Learning in Medical Education. *Npj Digital Medicine*, *8*, 627. https://doi.org/10.1038/s41746-025-02022-1

Zhang, J., Zhao, B., Yang, W., Foerster, J., Clune, J., Jiang, M., Devlin, S., & Shavrina, T. (2026). Hyperagents. *Arxiv Preprint Arxiv:2603.19461*. https://doi.org/10.48550/arXiv.2603.19461